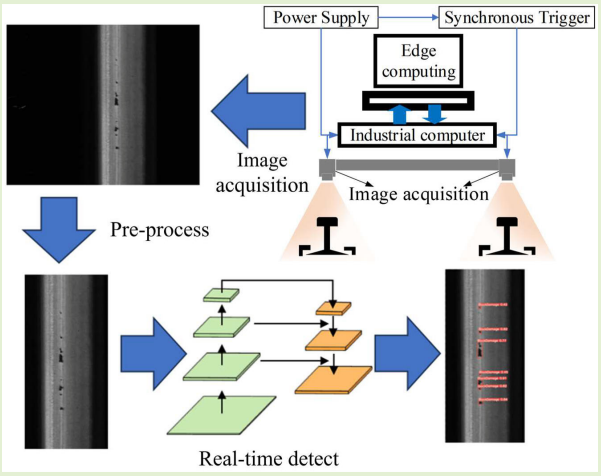
Research on Real-Time Detection System of Rail Surface Defects Based on Deep Learning

Wang Yaodong, Yu Hang, Guo Baoqing, Shi Hongmei, and Yu Zujun

***Abstract*—The heavy workload of rail track inspection makes it time consuming, and thus calls out a real-time inspection algorithm to achieve precise and efficient detection. In this study, we developed a real-time detection system for rail surface. Our system utilizes machine vision and real-time algorithms to ensure efficient and fast inspections. Edge computing device is used for real-time detection of track defect. To increase detection accuracy and speed, we optimized the YOLOv5 structure by introducing depth-separable convolution and reparameterization methods. Through training and evaluating the model on a dataset of rail surface defects, we achieved a mean average precision (mAP) of 83.2% and a detection speed of 51 FPS on edge computing devices. The performance of model outstrips that of other one-stage algorithms and backbone network detection results, as it exhibits high accuracy and speed. This achievement lays the groundwork for realizing real-time detection of rail defects and augmenting railroad safety.**



***Index Terms*— Deep learning, defect detection, machine vision, rail surface, real-time detection system.**

I. Introduction

RAILWAY infrastructure is crucial to the transportation sector. With the ongoing advancements in railway engineering, rail transport has become a progressively significant part of daily life. Accordingly, detecting rail damage has garnered increasing social attention. The speed and load of trains are escalating, leading to more regular interaction between train and rail. Coupled with the complex and changing environmental impacts, the lifespan of rails deteriorates [1], [2]. Thus, efficient, real-time, intelligent dynamic monitoring technology for rail is of utmost importance.

With the development of technology, many detection means are gradually applied to rail defect detection. At present, in addition to visual inspection, the commonly used means include eddy current [3] and ultrasonic [4] inspection methods. These methods can detect internal rail defects, but the detection results are not intuitive and often require expensive detection equipment. Moreover, the use of deep learning and other related research on detection and classification tasks has gradually become mainstream [5], [6], [7], [8], [9]. Nowadays, image acquisition systems and neural network methods have been applied to the railroad field [10], [11], [12], [13]. Image-based rail defect detection has likewise become a research hot spot, but achieving real-time detection, to identify defects while collecting image, is a challenge.

A. Related Work

Currently, detection methods for rail damage relies predominantly on manual intervention whereby workers walk along the railway to visually observe or knock to ascertain the extent of rail defects. Manual detection relies heavily on workers' technical experience, and its high labor intensity and overall inefficiency have made it increasingly difficult to meet current detection needs. In recent years, there has been a gradual shift toward the use of machine vision and image processing technology for rail surface defect detection. A rail defect detection system based on machine vision and morphological algorithms has been developed by Min et al. [14] and Liu and Wei [15]. Ni et al. [16] presented an edge feature-based algorithm for identifying rail defects to enhance detection

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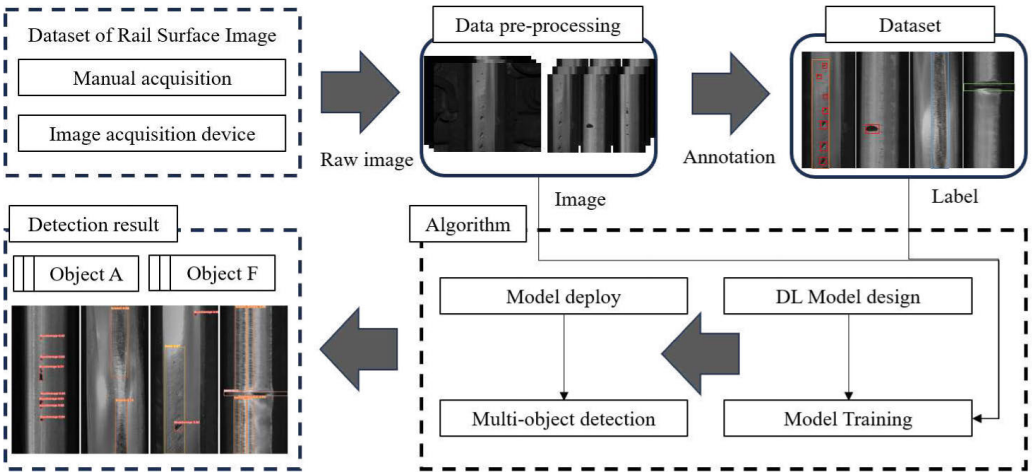


Fig. 1. Overall flow of the research.

accuracy. Similarly, Yu et al. [17] proposed a coarse-to-fine detection algorithm aimed at improving detection accuracy. In addition, Zhang et al. [18] introduced an automatic visual detection method that utilizes curvature filtering and Markov random field for rail defect segmentation.

Image processing-based rail defect detection quickly and accurately segments defects on the rail surface. However, the manual design of feature extractors can lead to poor robustness and detection accuracy, which is greatly impacted by environmental factors [19]. Deep learning-based defect detection methods have better scalability and application prospects when compared to image processing technology. Convolutional neural networks (CNNs) have been applied to image recognition, including the use of AlexNet [20], VGG [21], [22], ResNet [23], MobileNet [24], [25], [26], and others for image classification. However, rail images are often intricate, and sections of rail images can contain multiple defects that are challenging to detect through simple classification. Some scholars have used deeplab [27], U-net [28] and its upgraded network [29] for detecting image segmentation. Although this kind of neural network can classify and identify defective images at pixel level, it is challenging to determine the location of rail defects.

Two classes of deep learning object detection algorithms are widely used for Rail defect detection. One is the two-stage detection algorithms such as the RCNN series algorithms [30], [31], [32], [33], which use steps to identify and locate the objects. This type of algorithm has higher detection accuracy but slower speed. The other is one-stage detection algorithms, which can directly classify and locate defects for identification. Such as the widely used SSD algorithm [34] and YOLO series algorithms [35], [36], [37], [38], [39]. These algorithms tend to have high detection speed, but detection accuracy is somewhat lacking.

Mi et al. [40] improved the YOLOv4 algorithm for rail defect detection for small targets of rail defects. Anwar et al. [41] used the improved CSPDarkNet as a YOLOv4 feature extraction network to enhance the rail defect detection model. Guo et al. [42] and Bai et al. [43] improved the YOLOv4 algorithm using MobileNet to improve the detection speed of the algorithm model. Ni et al. [44] proposed a rail defect detection network based on the consistency of the estimated center point of the IoU guide. Wang et al. [45] used the M-ASFF and V-CBAM to improve the YOLOv5 network and enhance the detection accuracy of rail defects. Guo et al. [46] introduced Transformer into YOLOv5 to enhance defect detection. Zhang et al. [47] fused two subnetworks, SSD and YOLOv3, for rail defect detection. Zheng et al. [48] and Guo et al. [49] used Mask RCNN for the detection of surface defects on rails. Li et al. [50] used multibackbones as feature extraction network to improve the detection accuracy. The complex detection algorithm can accurately categorize the defects and identify their location. However, the amount of railway inspection data is huge, and in order to achieve fast detection, expensive equipment is often required.

B. Our Contributions

Existing research has shown that deep learning object detection algorithms can achieve a high level of accuracy in detecting defects on rail surfaces. At the same time, in the actual application process, there are also the following problems to be solved.

1) Models are usually too large to deploy on edge computing platforms, and take a long time to detect defects. It is difficult to achieve real-time detection to meet the actual engineering needs.
2) The current lightweight detection algorithms feature extraction capability is insufficient. It is difficult to meet the high-speed and high-precision detection requirements for rail defect detection.

To solve the above problems, this article designs and improves the YOLOv5 algorithm for the need for rapid detection of rail surfaces. For the characteristics of rail detection, we adjusted the model structure to reduce redundant calculations, combined with the depth separable convolution and structural reparameterization module [51] to improve the YOLOv5 algorithm, so that it can achieve the rapid detection of rail surface defects. Fig. 1 demonstrates the overall flow

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Algorithm 1 Model Reparameterization

Input: Dataset D, Model M
Dataset prepare:
for all $x \in D$ **do**
  $(x, y) \leftarrow$ preprocess(x) ▷ Resize and normalize
end for
Training Loop:
while not convergence or maximum iterations not reached **do**
  for all $(x, y) \in D$ **do**
    Get prediction: $y_{pred} \leftarrow M(x)$
    Calculate loss: $l \leftarrow$ Loss_function(y_{pred}, y)
    Update model: $M \leftarrow$ optimizer$(learn_rate, l, M)$
  end for
end while
Testing/Validation Stage:
$M_{rep} \leftarrow$ Reparameterize(M)
for all $x_{test} \in D_{test}$ **do**
  Predict output: $y_{pred} \leftarrow M_{rep}(x_{test})$
  Compute and accumulate performance metrics
end for
Output: Final model M_{rep} and performance evaluation results

of this research method, and Algorithm 1 shows the pseudo code of our work. The main contributions of this study are as follows.

1) A modular high-speed and high-precision image acquisition and inspection system is designed. With laser light source and line array scanning camera, it is capable of acquiring high-definition rail surface images.
2) An improved YOLOv5 algorithm is proposed, which reduces redundant calculation of the model and modifies the number of anchors. We introduce a reparametrized depth-separable convolution bottleneck module to improve the detection speed and detection accuracy.
3) The improved YOLOv5 model is compared with other models, and the dataset and RSDD dataset are utilized for experiments to fully prove the detection effect of the model. Meanwhile, the model detection speed is tested on edge computing devices to examine real-time detection.

C. Outline of the Article

In Section II of this article, we introduce the image acquisition equipment and the rail surface image dataset; In Section III, we describe the optimization methods of the algorithm model; In Section IV, we verify the conclusions of this article through experiments.

II. Hardware Systems and Datasets

A. Image Acquisition System

The image acquisition system is the key part in this study. We designed and fabricated a modular image acquisition system and mounted it on the inspection platform. As shown

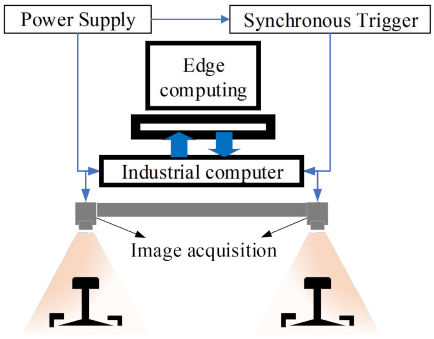


Fig. 2. Track image acquisition system. The system is composed of edge computing equipment, industrial computer, power supply, trigger, and image acquisition equipment.

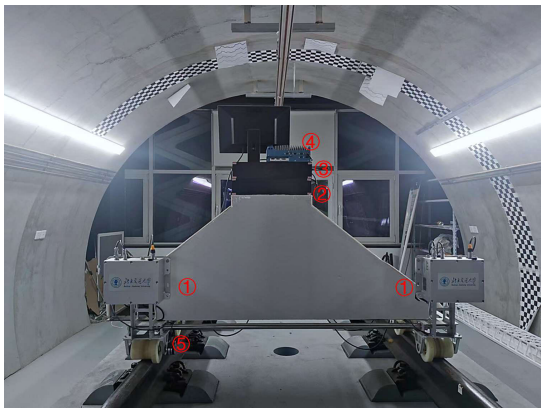


Fig. 3. Image acquisition platform. ① Image acquisition device. ② Power supply module. ③ Synchronized control module. ④ Industrial computer. ⑤ Rotary encoder.

in Fig. 2, the rail image acquisition hardware system mainly consists of image acquisition module, power supply module, synchronized touch module, industrial computer, and edge computing module and is integrated on the operation platform to form a set of acquisition system.

The image acquisition module parameters are calculated according to the maximum acquisition width T_w is 500 mm and acquisition accuracy R_f is 0.25 mm/pixel, then the required number of pixels R_i is

$$R_i = \frac{T_w}{R_f} = 2000 \text{ pixel}. \tag{1}$$

The image element size S is chosen to be 7.04 μm and the focal length of the lens F to be 12 mm, then the object distance u is

$$u = \frac{F \times R_f}{S} = 340 \text{ mm}. \tag{2}$$

Acquisition equipment, as shown in Fig. 3, consists of two Teledyne DALSA's CMOS line scan cameras with a resolution of 2048 pixels, using a 12 mm lens and a maximum line frequency of 52 kHz, with a laser light source for each camera to improve the image acquisition quality. The two cameras acquire the surface images of two rails separately. If the two cameras are triggered at a fixed frequency, the longitudinal distance between the pixels will change when the vehicle speed changes, which will affect the quantitative analysis of

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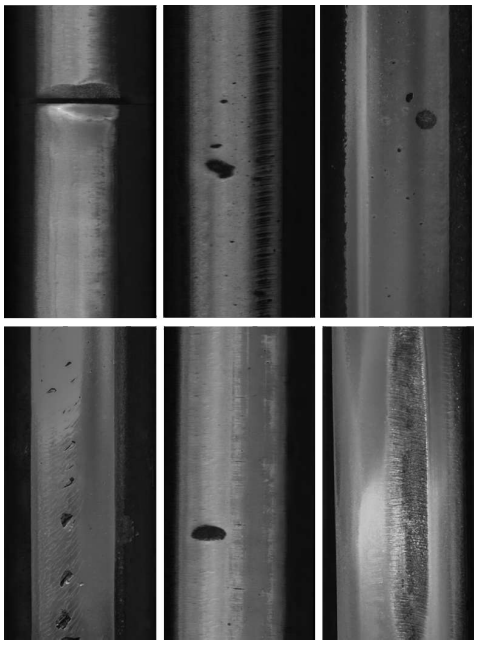


Fig. 4. Our dataset contains a variety of rail images. The data came from heavy carrier railways and subways.

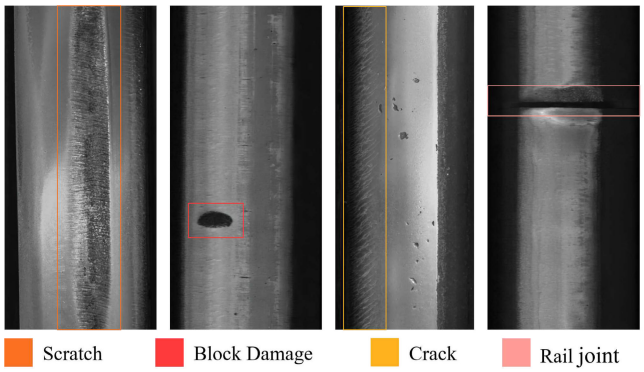


Fig. 5. Annotations of defects labeled in the dataset. Our dataset labels four kinds of target.

the geometric parameters of the rail surface defects. Therefore, in this study, a synchronized touch module is designed to match the acquisition frame rate and driving speed by encoder pulses. Pulses signals will be input into the synchronized touch module, and sending the processed TTL signals to the image acquisition device to control the image acquisition.

To achieve real-time detection processing at the edge, NVIDIA Jetson Xavier NX is used as the edge computing processor in this study. Compared to personal computer and server GPUs, these devices consume less power and are smaller in size. However, the limited arithmetic power makes it difficult to run complex algorithms, so a lightweight design for the algorithmic models is required.

B. Dataset

We have constructed a rail surface image dataset comprising over 2000 images. To enhance robustness, this study incorporates both urban rail data and heavy-haul railway data in the dataset shown in Fig. 4, which is collected by the acquisition deceives. A total of four categories are identified, namely rail surface scratch, crack, block damage, and rail joint (as shown in Fig. 5). The labeling in Fig. 6 displays information on the number, size, and location of defects in dataset.

The data are divided into a training set, a validation set, and a test set in a ratio of 7:2:1 for robust enhancing. To further verify the robustness of the algorithmic model, the RSDD dataset is also used for performance testing alongside the dataset we compiled.

TABLE I
ANCHORS IN EACH SCALE

Detect scale	Anchor size
1/32	(34,186), (34,534), (65,597)
1/16	(21,23), (53,39), (293,21)
1/8	(8,5), (10,8), (16,12)

III. IMPROVED YOLOV5

A. YOLOv5 Baseline

Real-time detection needs to detect quickly, yet accurately, and the model size must be constrained by edge detection equipment. Compared to other one and two-stage target detection algorithms, the YOLO series algorithms have a more balanced approach to the trade-off between speed and accuracy. Considering the accuracy, detection speed, and model volume for target detection algorithms, the YOLOv5s has been selected as the benchmark algorithm for multitarget detection on the track surface. The YOLOv5 algorithm, illustrated in Fig. 7, comprises three parts: CSPDarkNet, PA-FPN, and YOLOHead.

B. Detect Scale Adjustment

The YOLO series of algorithms uses multiscale detection, using K-means algorithm clustering to get nine anchors, three anchors are grouped into one group, and the three groups of anchors are assigned to large-, medium-, and small-scale feature maps, respectively, which are used for detecting large targets, medium-scale targets, and small targets, respectively.

In the rail defect dataset, the concentration of the target scale distribution is higher. The clustering algorithm has produced nine anchors, which are illustrated in Table I. These anchors have been assigned to three feature layers, forming three groups. The hierarchical anchor mechanism enforces hierarchical targets, which leads to some anchors with minor differences being detected in different feature layers. In the track surface defect dataset, larger targets such as joints, scratches, and cracks typically have elongated shapes with extreme aspect ratios, making them difficult to detect in small-scale feature layers. Due to the anchor grouping and layering mechanism in YOLOv5, the detection needs of the track surface defect dataset are challenging to satisfy. Consequently, some targets may remain undetected in more appropriate feature layers.

We adapted the YOLOv5 detection model, as illustrated in Fig. 8, to simplify the overall model architecture. Feature map outputs at 1/32 and 1/16 scales have been omitted, with emphasis solely placed on the detection of large-scale feature

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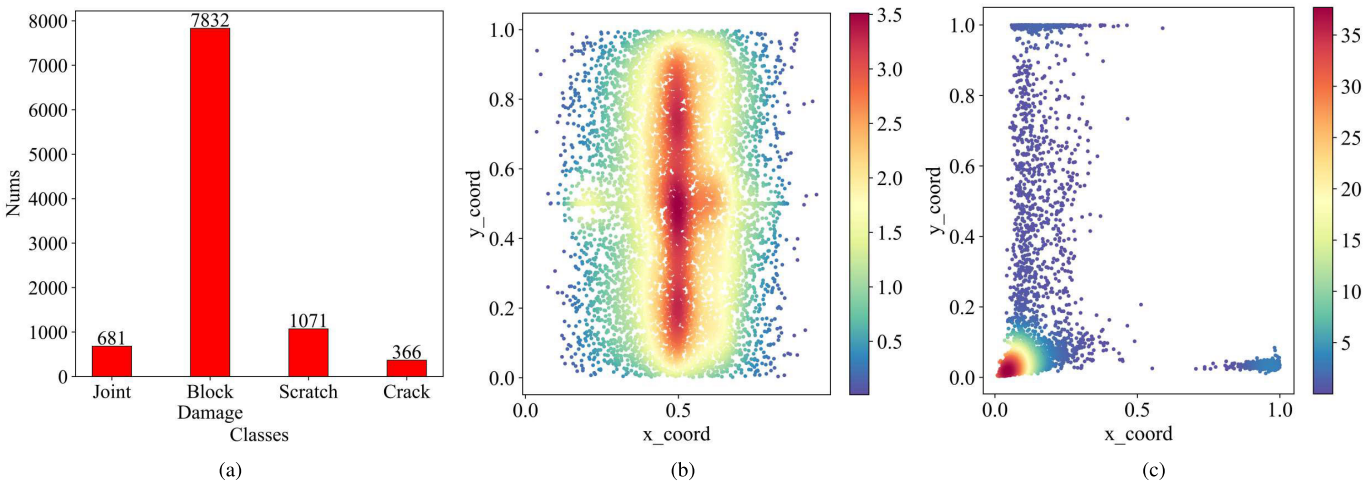


Fig. 6. Statistics for the target category, location, and size of the labels in the dataset. (a) Number of each category. (b) Object locations. (c) Object size.

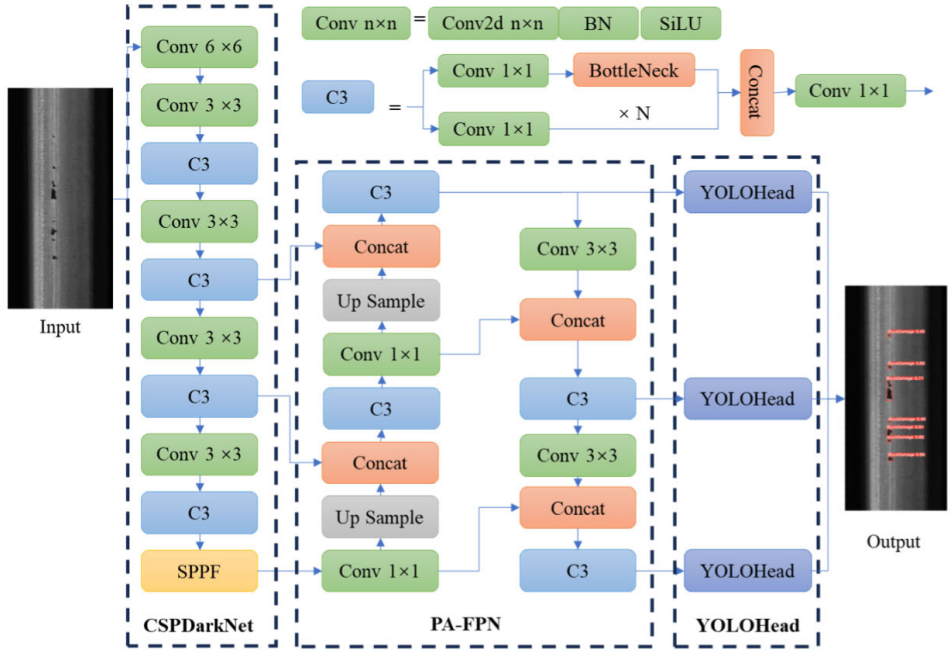


Fig. 7. YOLOv5 baseline.

maps. As smaller detection feature maps are unnecessary, the PA-FPN framework is streamlined and adjusted to the FPN structure for feature fusion purposes.

C. Anchor Resetting

Eliminating branching structures results in fewer anchors and consequently fewer model predictions, which may have a slight impact on the model's performance. Given the varying target distribution scales between the dataset of rail surface defects and that of natural scenes, the model's maintenance of the initial setting of nine anchors becomes somewhat redundant. Moreover, using too many anchors would increase computation during prediction result postprocessing, including feature map decode and nonmaximum value suppression, thereby negatively impacting detection speed. The number and size of anchors for the target detection algorithm were redefined using the K-means clustering algorithm. Clustering results are shown in Fig. 9, with three, five, and nine anchors.

D. Model Lightweight

Edge computing devices have a constrained storage capacity, and deploying large models to them is a challenge. Additionally, the limitations in both chip power and arithmetic power make it arduous for such devices to conduct fast and intricate calculations. Consequently, the real-time detection system for rail defects necessitates a lightweight algorithm model ton afford elevated efficiency and computation power.

Depth-wise separable convolution, as illustrated in Fig. 10, is a frequently applied method in convolutional neural networks to decrease the model's parameters and computation. It decreases the computational workload and parameter count

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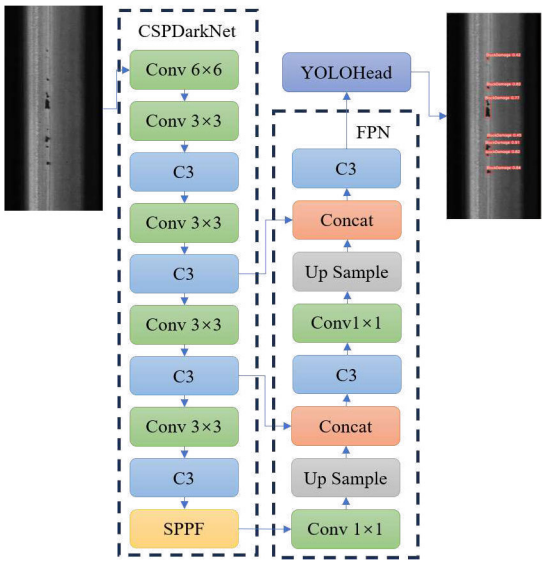


Fig. 8. Adjusted YOLOv5. Structure of the model is simplified.

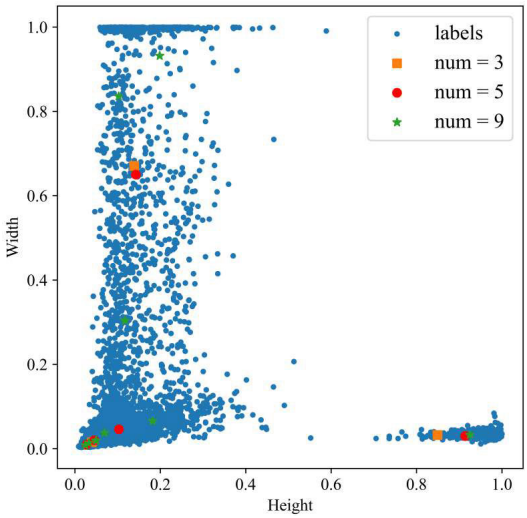


Fig. 9. Dataset class clustering results with different anchor number.

by dividing the input channel into multiple subchannels and executing a separate convolution operation on each one. On the basis of channel convolution, point-wise convolution operations is applied, blending the outcomes of distinct subchannels, to get the final feature map. Replacing traditional convolution with depth-separable convolution in the algorithmic model can significantly decrease the computational volume and model size, thus enhancing the detection speed considerably. The YOLOv5 model replaces the bottleneck module with the depth-wise separable convolution bottleneck (DWB) module achieved through depth-wise separable convolution instead of conventional convolution. The basic structure of the bottleneck and DWB module is illustrated in Fig. 11(a) and (b).

Although depth-wise separable convolution reduces the volume and computation of the model, which significantly enhances the detection speed, it consequently leads to a decrease in model complexity and overall accuracy. To address this issue, the method of reparameterization is proposed

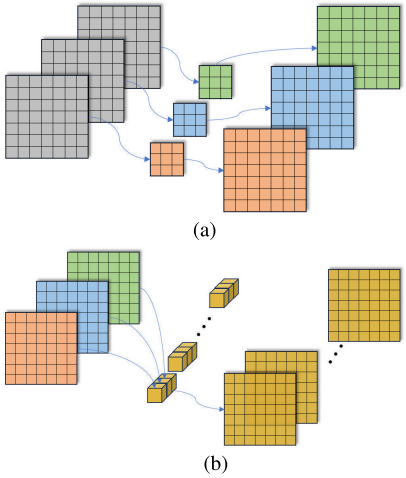


Fig. 10. Depth-wise separable convolution and point convolution. (a) Depth-wide convolution. (b) Point-wide convolution.

for improving detection accuracy while maintaining the model volume and computation amount unchanged during deployment and inference. The structure of the proposed reparameterization DWB (RDWB) using structural reparameterization is shown in Fig. 11(c). Add a 1 × 1 convolution and an additional 3 × 3 deep-wise convolution to the basic 3 × 3 deep-wise convolution in each DWB module during training. In the deployment inference stage after the model training, the three branches will be merged to ensure the detection speed while improving the model accuracy.

The convolution operation is typically linked to the BN layer. During the reparameterization process, the batch normalization layer is initially merged with its prior convolution layer to generate a biased convolution block, the process can be described as the following equation:

$$\begin{aligned} \mathrm{BN}(x) &= \gamma \frac{x-\mu}{\sqrt{(\delta^2+\epsilon)}} + \beta \\ \mathrm{BN}(W \otimes x) &= \gamma \frac{(W \otimes x-\mu)}{\sqrt{(\delta_i^2+\epsilon)}} + \beta \\ W' &= \frac{\gamma \cdot W}{\sqrt{\delta^2+\epsilon}} + \beta \\ \beta' &= \frac{-\gamma \cdot \mu}{\sqrt{(\delta^2+\epsilon)}} + \beta \\ \mathrm{BN}(W \otimes x) &= W' \otimes x + \beta' \end{aligned} \tag{3}$$

where x is an input feature map; BN is a batch normal layer. γ, β, μ and δ are scaling factor, bias factor, mean, and variance of BN Layer. $\otimes$ is convolution operation, and W is weight of convolution kernel. W' and β' are weight and bias of new convolution operation.

Weights of 1 × 1 convolution can be viewed as a 3 × 3 convolution with 0 at edges and 1 × 1 convolution weights in the center. Therefore, all branches can be converted to 3 × 3 convolutions. By adding the weights and biases of the equivalent 3 × 3 convolutions obtained in all branches, they can be merged into a single one. The conversion of structural

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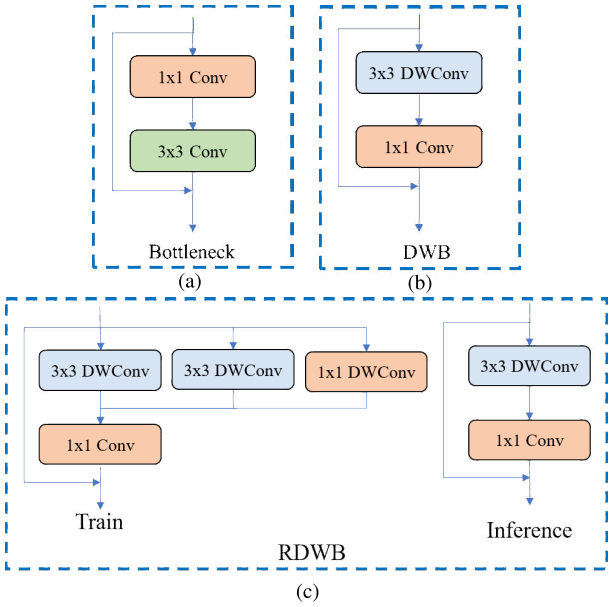


Fig. 11. Structures of initial bottleneck, DWB block and RDWB block. RDWB modules use complex structures during training and simpler equivalent structures during inferencing. (a) Initial bottleneck. (b) DWB. (c) RDWB.

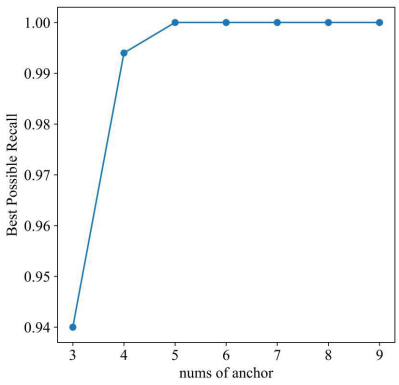


Fig. 12. Best possible recall in different numbers of anchor.

reparameterization can be expressed as following equation:

$$\begin{aligned} F &= W_3'^1 \otimes x + \beta_3'^1 + W_3'^2 \otimes x + \beta_3'^2 + W_1'^1 \otimes x + \beta_1'^1 \\ &= \left(W_3'^1 + W_3'^2 + W_1'^1\right) \otimes x + \left(\beta_3'^1 + \beta_3'^2 + \beta_1'^1\right) \\ &= W' \otimes x + \beta' \end{aligned} \tag{4}$$

where F is the output feature map. $W_i'^j$ and $\beta_i'^j$ means the weight and bias of the jth convolution kernel of size $i \times i$.

IV. Experiment Result

Experiments used mAP at β (mAP@β) as an evaluation index. Models prediction results were compared with real annotation results, and the results with the IoU greater than threshold β were marked as true positive (TP), the results less than β were marked as false positive (FP), and the results not detected by models were marked as false negative (FN). The precision and recall are calculated by the following equations. The area integral of the plotted P-R curves was AP, and the mean of all detected classes is mAP

$$\begin{aligned} \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}}. \end{aligned} \tag{5}$$

First, we verified the performance of model with the different detection scale selections. Models use CSPDarkNet as the backbone for feature extraction, and FPN for fusing features at different scales. Based on the maximum possible recall rate obtained from clustering, as shown in Fig. 12, the number of anchors was selected to be 5. The feature maps with the scales of 1/4, 1/8, 1/16, and 1/32 were used to carry out the training and detection. Table II shows the experiments' result.

The best detection results of 83.2% were obtained at the detection scale of 1/8, with a difference of about 4% in the category of cracks compared to the other scales. Joints, which

TABLE II
Comparison of Different Detection Scale

Detect scale	mAP@0.5	AP@0.5				Complexity (GFLOPs)
		Joint	Block Damage	Scratch	Crack	
1/4	0.818	0.995	0.805	0.635	0.838	14.5
1/8	0.832	0.995	0.768	0.664	0.900	12.9
1/16	0.825	0.995	0.773	0.675	0.859	11.4
1/32	0.802	0.995	0.728	0.629	0.855	10.7

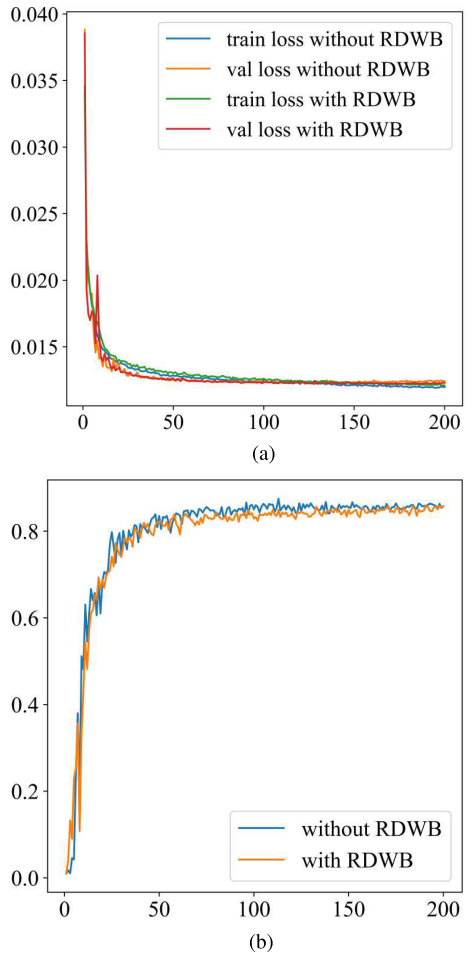


Fig. 13. Loss and mAP@0.5 for each epoch of our model during training and validation. (a) Loss. (b) mAP@0.5.

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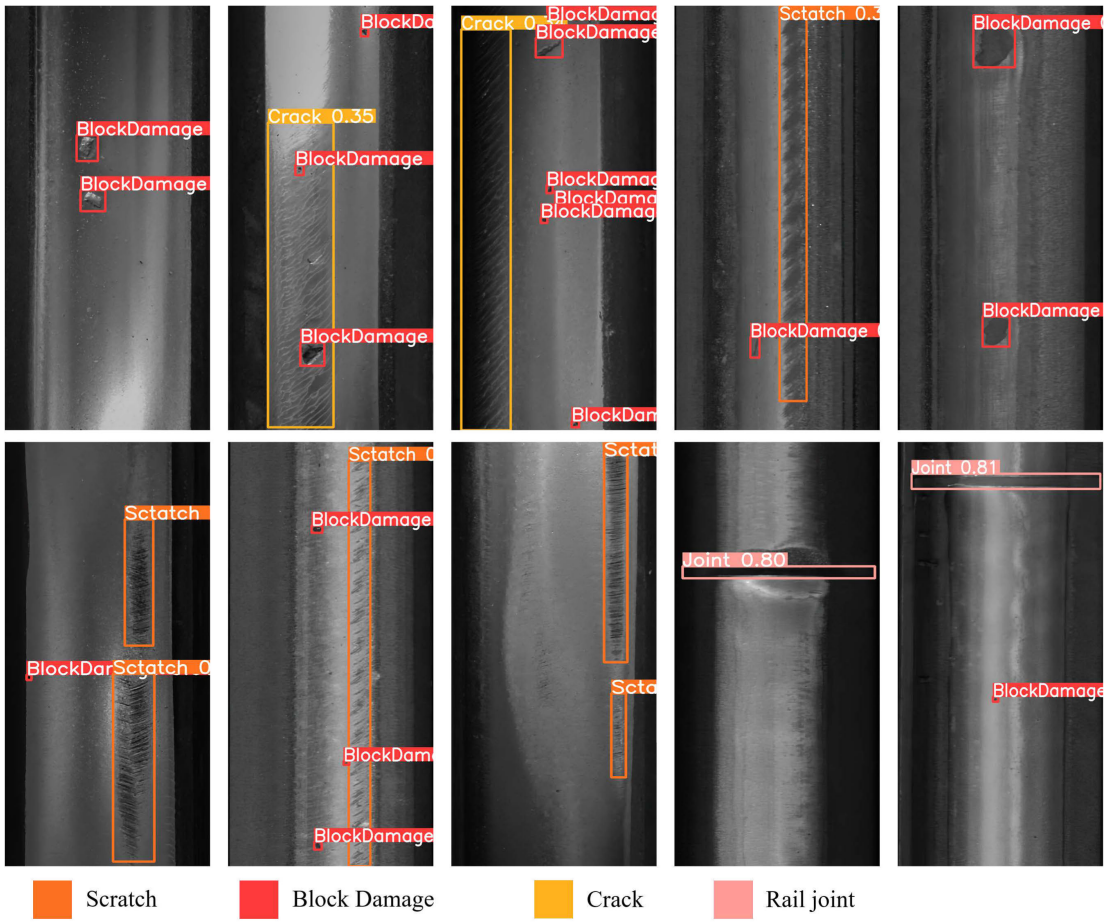


Fig. 14. Detect results of proposed model can show the performance of the model more intuitively.

TABLE III
COMPARISON IN DIFFERENT NUMBERS OF ANCHOR

Numbers of anchor	mAP@0.5	AP@0.5				Complexity (GFLOPs)
		Joint	Block Damage	Scratch	Crack	
3	0.820	0.995	0.766	0.651	0.868	12.9
4	0.818	0.995	0.778	0.645	0.855	12.9
5	0.832	0.995	0.768	0.664	0.900	12.9
6	0.819	0.995	0.773	0.659	0.848	13.0
7	0.819	0.995	0.772	0.631	0.878	13.0
8	0.832	0.995	0.743	0.695	0.894	13.0
9	0.830	0.995	0.770	0.685	0.867	13.0

TABLE IV
COMPARISON OF DIFFERENCT LIGHTWEIGHT STRUCTURE

Backbone	mAP@0.5	F1 score	Complexity (GFLOPs)
DWB	0.802	0.795	8.8
RDWB	0.824	0.801	8.8

are relatively fixed in shape and have distinctive features, have a better detection effect, reaching 99.5% in all four inspection scales. As the inspection scale is gradually reduced, the model's detection effect for small targets decreases. As both scratches and cracks are extreme aspect ratio targets, their widths are difficult to match the feature map for detection while the detection scale is gradually reduced, and thus the detection effect gradually decreases. At the same time, due to the complexity and variability of the feature of abrasion, and easy to be confused with the background, resulting in all scales on the detection effect indicators are lower than the rest of the categories. Considering the amount of computation, using a 1/8 scale for detection can get a better detection effect.

Table III shows that the different number of anchors has a certain impact on the performance of the model. With the increase in the number of anchors, the overall detection effect shows a gradual increase in the trend. When the number of anchors is greater than five, the overall mAP tends to converge. When more anchors are used, the computational amount of the model will slightly increase, and the prediction results output will also increase. Too many prediction results will increase the computational complexity of the postprocessing, which in turn affects the model detection efficiency. Therefore, considering the comprehensive detection effect and detection accuracy, the number of anchors is selected as five.

We compared the detection results before and after using the structural reparameterization module in Table IV. Using the RDWB module to construct the algorithmic model, the computation of the model during inference process is still 8.8 GFLOPs, but the mAP is improved by 2% compared to using the RDWB model. Fig. 13 shows the loss and mAP curves of the proposed models during training.

To further validate the effectiveness of the model, the proposed model is compared with the better target detection models of recent years. Table V shows the detection results of these models, which in turn validates the effectiveness of

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TABLE V
COMPARISON OF DETECTION RESULTS USING DIFFERENT MODELS

Model		mAP@0.5	AP@0.5					Complexity
			Joint	Block	Damage	Scratch	Crack	(GFLOPs)
Baseline	YOLOv4	0.795	0.995	0.742		0.643	0.800	20.6
	YOLOX	0.813	0.995	0.758		0.621	0.877	21.6
	YOLOv5	0.825	0.995	0.769		0.634	0.900	15.9
	YOLOv7-tiny	0.787	0.995	0.763		0.585	0.806	13.9
	Ours	0.832	0.995	0.768		0.664	0.900	12.9
Lightweight	ShuffleNet	0.792	0.995	0.769		0.577	0.829	7.6
	MobileNet	0.795	0.995	0.741		0.604	0.839	8.0
	DWB	0.810	0.995	0.77		0.668	0.806	8.8
	RDWB	0.824	0.995	0.786		0.666	0.848	8.8

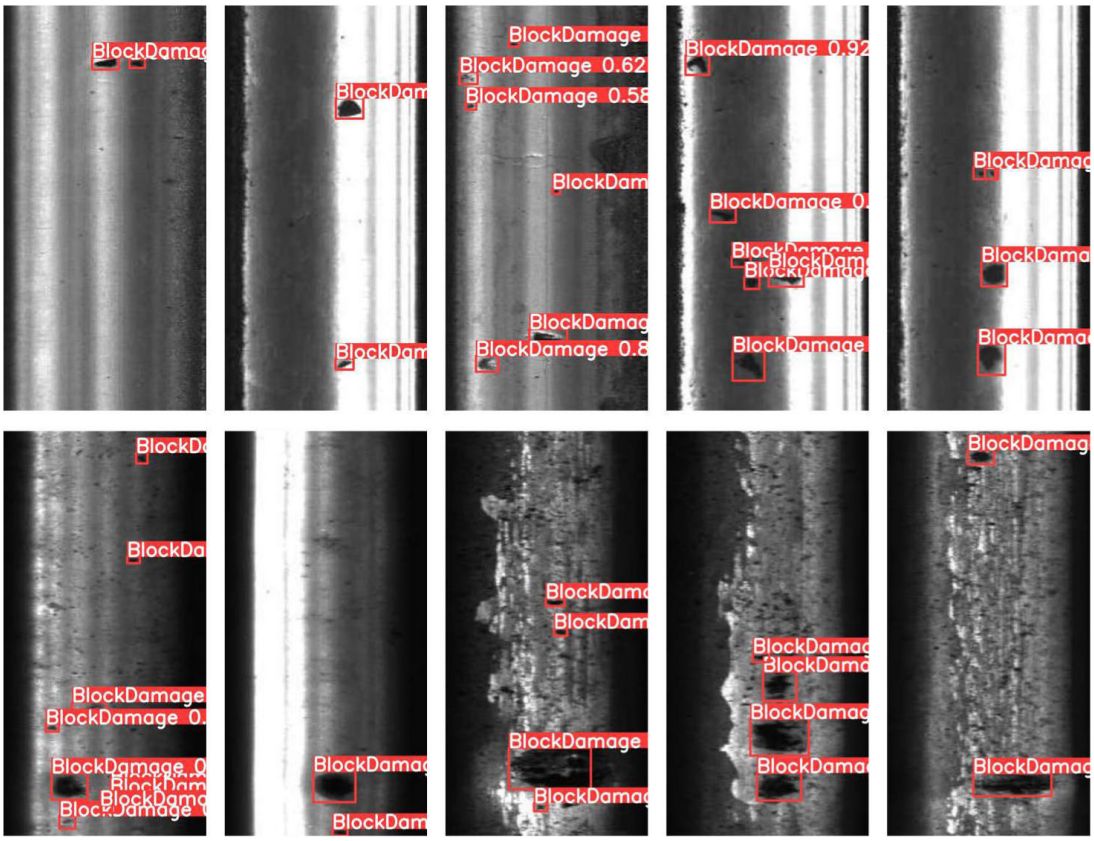


Fig. 15. Detection results of the model in RSDD dataset can intuitively show the generalization performance of the model.

TABLE VI
COMPARISON OF DETECTION SPEED USING DIFFERENT MODELS

Model	Complexity (GFLOPs)	Time(ms)	FPS
YOLOv5s	15.8	26.8	37.3
Ours	12.9	22.4	44.6
Ours + RDWB	8.8	19.6	51.0

the models on this dataset. Fig. 14 presents selected detection outcomes of the proposed model, which demonstrate that it can effectively detect most targets. However, some small and featureless targets were still subject to misdetection and omission.

The algorithm proposed in this article is compared with YOLOv4, YOLOX, YOLOv5, and YOLOv7-tiny. Its detection mAP is improved by 3.7%, 1.9%, 0.7%, and 4.5% and the computation complexity is decreased by 7.7 GFLOPs, 8.7 GFLOPs, 3 GFLOPs, and 1GFLOPs, respectively. Meanwhile, lightweight reconstruction by using RDWB leads to an improvement of the model's accuracy concerning that of using ShuffleNet and MobileNet by 3.2%–2.9%, respectively. The amount of computation only increases by 1 GFLOPs. The above experiments fully validate the effectiveness of the model.

We further verify the model's effectiveness and the dataset's representativeness through a set of robust experiments on the RSDDs dataset which is not included in our dataset. The dataset images have been cropped to ensure that each image's aspect ratio aligns with that in the track surface defects dataset. By using the model proposed in this article, the target detection results are shown in Fig. 15, demonstrates the model's ability to complete the detection experiments in this dataset. Despite missing a few very small targets, the model effectively detects most defects in the image and carries out basic track surface defect detection. The experimental results

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illustrate that the proposed model in this research and the constructed dataset exhibit excellent robust performance and are suitable for application in similar scenarios.

Table VI demonstrates inference speed of the models in this article, which takes 22.4 ms to deploy a single frame detection without RDWB model on the edge computing device and 19.6 ms with RDWB module.

V. Conclusion

In this study, we propose a rail surface real-time inspection system, which can realize multiplatform migration through modular design and can complete the automatic acquisition of multichannel rail surface images.

We improve the YOLOv5 algorithm model to streamline the algorithm structure, reduce the amount of redundant calculations, and increase the detection speed. To achieve real-time detection on NVIDIA Xavier NX, the RDWB module is proposed based on the idea of structural reparameterization, and the model is designed to be lightweight. So that the model on which the detection speed finally reached 51 FPS, the detection speed, can meet the demand for real-time detection in low-speed situations.

In future work, we plan to collect rail defect data as comprehensively as possible to study more complex rail defects. Additionally, we will explore advanced techniques such as unsupervised detection methods and domain adaptation detection to enhance the performance and efficiency of the detection algorithm.

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